Songyu Ye

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Education

University of Southern California Aug 2025 - Dec 2026 expected

M.S. in Analytics | GPA: 3.767/4.0

The Ohio State University Aug 2021 - May 2025

B.S. in Computer Science; Minor in Communication Technology | GPA: 3.67/4.0

Research Interests

World models and medical world models; probabilistic modeling of disease progression; image forensics and artifact representation learning; robust clinical prediction with tabular foundation models.

Research Experience

Publications

[1] **NeurIPS 2026** | Main Track Oral | **Top 0.4% of valid submissions**

Oral selection: 112 of 30,709 valid submissions (~0.4%). Official listing | Statistics

X. Zhu, K. Feng, R. Wang, X. Wang, X. Ma, B. Du, C. Jiang, C. Qu, **Songyu Ye**, X. Du, W. Feng, J. Liu, and J.-Z. Zhou. *Can We Model the Artifacts Explicitly? Disentangle Artifacts via Pairwise Edit Relations for Image Manipulation Localization.*

[2] **ICONIP 2026** | CCIS proceedings | Accepted

Songyu Ye, M. Pishgar, M. Chen, Y. Chen, S. Zhao, X. Du, Z. Lin, and J.-Z. Zhou. *Recall First ARDS Prediction via Inductive Bias Diversity in CatBoost and TabPFN Ensembles.*

[3] **ICONIP 2026** | LNCS proceedings | Accepted

J. Xu, K. Feng, J. Lv, Y. He, and **Songyu Ye**. *EAGARN: Edge Artifact Guided Autoregressive Network in Image Manipulation Localization.*

[4] **Journal of Imaging** | 12(10), 522 | Published 20 Oct 2026

Z. Lin, Z. Wang, T. Xu, **Songyu Ye**, X. Xu, and C. Peng. *SMICA: Semantic and Multi-Granularity Integration with Constrained Adaptation for Fake Image Detection.*

Paper result: 92.11% macro-average AUC on OpenMMSec across four domains, training 2.50% of parameters.

Research Experience continued

Research Projects

Medical World Models for Longitudinal Brain Tumor MRI In progress

- Investigated near-persistence collapse and weak sensitivity to follow-up intervals in longitudinal brain MRI forecasting; used decoder controls and semi-synthetic temporal tests to isolate transition-model limitations.
- Developed and evaluated change-conditioned latent flow models and an interval-structured mixture head; assessed calibration and progression predictions against persistence and state-only baselines.

Recall First ARDS Prediction 2025 - 2026

First Author and Independent Project Lead

- Formulated early warning as recall maximization under a precision constraint; designed weighted probability fusion of CatBoost and TabPFN to exploit complementary errors and recover missed ARDS cases.
- Analyzed false-negative overlap and compared loss baselines and fusion ablations to test model complementarity; evaluated robustness to missing data on MIMIC-IV. Led implementation, experimental interpretation and manuscript writing.

ForensicHub May 2025 - Sep 2025

Open Source Research Contributor | Unified Image Forensics Benchmark

- Bridged localization and classification interfaces by integrating mask-to-label adapters, pooling spatial responses into image-level scores and handling model-specific frequency inputs and output formats.
- Standardized preprocessing across forensic tasks and authored configuration-driven training and evaluation tutorials, supporting reproducible experiments beyond the original IMDLBenCo coverage.

Social Dispute Analysis and Resolution Technologies May - Aug 2022

Research Assistant | University of Science and Technology of China | Prof. Zuocheng Wang

- Analyzed 100+ dispute scenarios and consolidated questionnaire branching logic; structured response records for downstream dispute assessment and triage.

Technical Skills

Modeling: PyTorch; latent flow matching; probabilistic forecasting; CatBoost/TabPFN ensembles; artifact feature fusion.

Validation: Precision-constrained threshold selection; loss and fusion ablations; calibration analysis; patient-level bootstrap; failure analysis.

Medical imaging: 3D MRI preprocessing and registration; MONAI, NiBabel, SimpleITK.

Programming: Python, Java, JavaScript; Git; reproducible experiment configuration.

Teaching and Software Experience

Teaching Assistant in Software Engineering Jan - May 2024

The Ohio State University | Led programming labs, supported debugging and project development, and evaluated code quality in assignments and exams.

Dice1L Jan - May 2024

Designer and Developer, The Ohio State University | Built and tested a reusable Java dice-simulation component with configurable outcomes and interface integration.